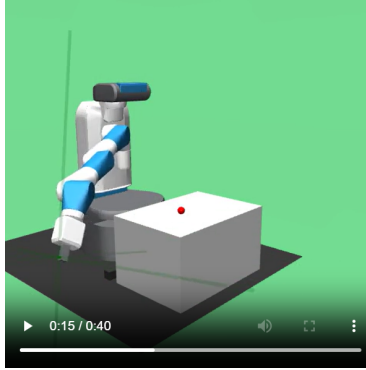


Effects of Enhancing TQC with Reward Shaping on Early Training Dynamics in FetchPickAndPlace-v4

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Abstract. In this work, we analyze the early training behavior of TQC in the *FetchPickAndPlace-v4* environment to answer two questions: (i) whether baseline TQC without Hindsight Experience Replay (HER) can learn effectively under sparse rewards, and (ii) whether reward shaping improves early training dynamics for TQC+HER. We compare baseline TQC, TQC+HER and reward shaping enhanced TQC+HER under identical training conditions and evaluate performance using success rate, mean episode return, area under the curve for the success rate and variance across random seeds reported as the standard deviation. Significance is evaluated via non-parametric tests. The results show that baseline TQC fails to learn in this environment, and also that augmenting TQC with HER does not yield improved learning. The results also show that there is no statistically significant improvement yielded from the reward shaping enhancement; suggesting that simple reward shaping on its own is insufficient to reliably improve early learning dynamics in sparse robotic manipulation tasks.

Keywords: Reinforcement Learning · Actor-Critic · TQC · Hindsight Experience Replay · Reward Shaping · Robotic Manipulation · Gymnasium-Robotics

1 Introduction

1.1 Background

While deep reinforcement learning has achieved strong performance in continuous control and robotic manipulation thanks to advances in off-policy actor-critic methods and function approximation [9, 6, 12], many state-of-the-art algorithms remain sensitive to environment characteristics, exhibiting overestimation bias [5] and unstable early training dynamics [7, 13].

The Truncated Quantile Critics (referred to as **TQC** hereafter) [8] model was introduced as an extension of distributional RL methods to address the overestimation bias. By combining quantile regression with conservative value estimation, **TQC** provides improved stability on continuous control benchmarks [8]. While prior work has largely focused on final performance metrics, less attention has been given to how enhancements to **TQC** influence *early training dynamics*, where learning instability and variance are most prominent [7, 13]. In robotic manipulation settings, early learning behavior is of particular importance [9]. These environments contain long-term dependencies, sparse rewards, and complex object interactions, making models sensitive to initial policy exploration [15]; with some techniques developed to mitigate these issues, for instance *Hindsight Experience Replay* (referred to as **HER** hereafter) [2].

1.2 Motivation & Goal

In this work, we study the *FetchPickAndPlace-v4* robotic manipulation task from the OpenAI Gymnasium robotics library, a standard benchmark for goal-conditioned continuous-control manipulation [2, 4, 16]. We investigate the following research questions:

- **RQ1:** To what extent can baseline **TQC** without **HER** achieve early-stage learning behavior in the *FetchPickAndPlace-v4* environment?
- **RQ2:** How does an enhanced variant of **TQC+HER** utilizing reward shaping compare to standard **TQC+HER** during early training in the *FetchPickAndPlace-v4* environment?

We focus on early training dynamics-learning speed, seed variance, and short-horizon performance by comparing **TQC**, **TQC+HER**, and an enhanced **TQC+HER** variant; as this better reflects the practical importance of training efficiency and stability in robotic manipulation [3, 7, 10].

2 Methodology

2.1 Algorithm

We use **TQC** [8] as the underlying reinforcement learning algorithm in all experiments. **TQC** represents the state-action value function via quantile regression and mitigates overestimation by truncating high-value quantiles across critic ensembles, resulting in a conservative value estimate with improved stability in continuous control tasks. We do not modify the **TQC** algorithm itself; the policy and critic architectures, loss functions, and update rules follow the original formulation. This ensures that observed differences in learning behavior, particularly during early training, can be attributed solely to the proposed enhancement [8].

2.2 Environment

We evaluate all methods on the *FetchPickAndPlace-v4* environment from Gymnasium-Robotics [4, 16], a widely used benchmark for goal-conditioned continuous control manipulation derived from the Fetch robot simulation [2]. In this task, a 7-DoF robotic arm must grasp an object from a table and place it at a target location in three-dimensional space [2, 14]. The environment is episodic and features sparse rewards, making effective exploration and stable value estimation particularly important during early training [5, 7]. Observations follow a goal-conditioned formulation and include robot states, object states, and the desired goal, while actions are continuous and control end-effector motion and gripper actuation [2].

2.3 Hindsight Experience Replay (HER)

While TQC can be applied directly to continuous-control tasks, prior work has shown that robotic manipulation environments such as *FetchPickAndPlace-v4* are particularly challenging due to sparse rewards and long-horizon dependencies. As a result, it is standard practice to combine off-policy actor-critic methods with HER to improve exploration and sample efficiency in these settings [2, 14]. In this work, we therefore consider HER as part of the standard training setup for *FetchPickAndPlace-v4*.

2.4 Reward Shaping

Our proposed reward shaping is applied on top of the TQC+HER configuration. First, a flat reward of 0.1 (a tenth of the completion reward) is introduced when the robot makes contact with the object. This is the first step of the picking and placing task. For this reason we chose to add this initial reward. Second, the agent is rewarded when it grips and lifts the object. This comprises both the gripping and lifting (which are the next sub-tasks), which is essential in this task. Finally, an incremental award is added to this holding reward. The closer the agent is to the target position, while holding the block, the greater the addition to the flat reward (it ranges from 0.4 to 0.6). The values of these sub-task rewards were selected with our knowledge of the domain in mind, which has its limitations [1]. The initial contact reward should not break the reward structure, but it is an important first step. The hold and moving closer reward is the most essential part of the task.

3 Experimental Setup

To provide a complete empirical analysis, we evaluate three variants: (i) TQC, (ii) TQC+HER, and (iii) TQC+HER with the proposed enhancement. We conduct two complementary experiments: first, TQC versus TQC+HER to assess the impact of goal relabeling in the sparse-reward *FetchPickAndPlace-v4* environment; and second, TQC+HER versus its enhanced variant, treating TQC+HER as the primary baseline in line with common practice for robotic manipulation tasks [2, 14]. All models are trained using identical environment configurations, network architectures, optimization procedures, and hyperparameters; and differ only in the applied reward shaping described in Section 2.3. Experiment 2 is repeated across seeds {10, 42, 55}. Training is performed for a fixed number of environment interaction steps (see Section 3.1).

Table 1. Hyperparameters used for all TQC models during training.

Hyperparameter	Value
Total training timesteps	100,000
Policy type	MultiInputPolicy
Learning rate	0.001
Batch size	2048
Replay buffer size	100,000
Discount factor (γ)	0.98
Target smoothing coefficient (τ)	0.05
Policy network architecture	[512, 512, 512]
Number of quantiles (N)	25
Top quantiles dropped per critic	2

Table 2. Hyperparameters used for HER variants during training.

HER Hyperparameter	Value
Replay buffer class	HER
Goal sampling strategy	Future
Sampled goals (n_{sampled})	4

3.1 Hyperparameters

All models use identical hyperparameters across all experiments, with additional HER parameters for respective variants, summarized in Tables 1 and 2. These values follow standard practice for TQC and continuous-control benchmarks [8]. Since our experiments focus on early learning behavior, limiting training to the first 1×10^5 timesteps is sufficient. We use a `MULTIINPUTPOLICY` to handle the structured observation space of the task, which includes gripper state, object information, and goal coordinates. The learning rate is chosen to balance stable optimization with the additional learning signal introduced by HER; although some works employ separate learning rates for the actor and critic [5], we avoid this to keep the experimental setup simpler. A large batch size is used to better exploit the dense information provided by HER during training, while the replay buffer size is set to 1×10^5 to allow reuse of all experiences. The discount factor (γ) ensures that later parts of the task remain sufficiently valued. The target smoothing coefficient (τ) is set relatively high to allow more frequent target network updates. A deeper policy network is used to capture the complex relationships between the gripper, object, and target, while a higher number of quantiles improves the representation of action-value uncertainty. Dropping the top two quantiles reduces overestimation, and using future goal sampling in HER provides sufficient goal diversity [2].

3.2 Evaluation Metrics & Statistical Testing

We evaluate the experiments using four metrics: *success rate*, which directly measures task completion in the Fetch manipulation setting [2, 4, 16]; *mean episode return*, which reflects learning progress before successful episodes become frequent in sparse-reward environments [13]; early-training *AUC* of the success-rate curve over timesteps, summarizing learning speed and early performance [7]; and *seed variance*, reported as the standard deviation across the seeds {41, 10, 55} to assess training stability. We use a non-parametric Mann-Whitney U test to compare the distributions of success rates between TQC+HER and Enhanced TQC+HER at each timestep. This test is appropriate due to the non-Gaussian, often binary nature of success rates in sparse-reward settings and the limited sample size, and allows us to assess whether observed performance differences are statistically significant [17, 11].

4 Results

Table 3. AUC and success rate results.

Algorithm	AUC	Success
TQC	0.0000	0.0000
TQC+HER	0.0167	0.0000
E. TQC+HER	0.0167	0.1333

Table 4. Statistical test result (p -value).

TQC+HER vs. E. TQC+HER
0.1876 ($p > 0.05$)

* AUC and success rates were computed using the evaluations from our trainings.

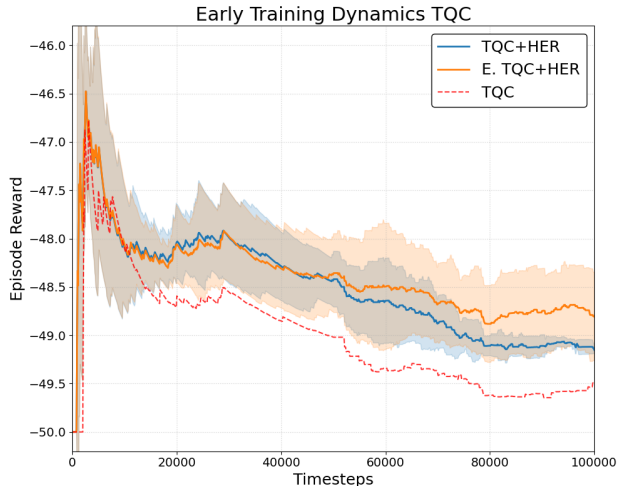
5 Discussion

The baseline TQC and TQC+HER models showed similar performance; however, no statistical testing was performed since TQC achieved zero success throughout training. The TQC baseline was therefore included primarily to demonstrate the relevance of HER in this environment [2], with being ran only on one seed (42) and not having standard deviations, further limited by computational and time constraints. All three algorithms exhibited similar mean episode return trajectories, with an initial increase likely driven by exploration [7], followed by a degradation in performance over time. Although the Enhanced TQC+HER variant achieved the highest final performance, the difference relative to TQC+HER was not statistically significant. Pointwise Mann-Whitney U testing revealed the lowest p -value at the final timestep, which nevertheless remained above the significance threshold. Due to limited training timesteps, sparse rewards, and coarse evaluation logging, success-rate measurements were sparse, noisy and not smooth [7], rendering the AUC metric uninformative: TQC attained an AUC of zero, while AUC values for the TQC+HER variants were extremely low and likely inflated again by exploratory behavior [7]. Seed variance analysis showed similar trends across both TQC+HER variants for most of training, except near the end where standard TQC+HER stabilized while the enhanced model did not. Pointwise statistical testing was preferred, since aggregation would collapse early training dynamics into a single scalar [11, 17, 7].

6 Conclusion

This study examined two research questions. First, baseline TQC without HER did not exhibit any early-stage learning and showed no difference from TQC+HER, contrary to prior work [2, 14]. Second, a reward shaping-enhanced TQC+HER variant did not yield statistically significant improvements in early training dynamics. This study was limited by compute and time constraints. Future work should consider more expressive reward shaping, additional random seeds for stronger statistical testing, and evaluations beyond early training.

Fig. 1. Mean episode return and standard deviations.



GitHub Repository. The full code of this work can be found in the GitHub repository, under the A2_RL_N_ROBOTICS folder:
 – <https://github.com/HippolyteLC/Project-Situated-AI-2026>

References

1. Andrychowicz, M., Wolski, F., Ray, A., Schneider, J., Fong, R., Welinder, P., McGrew, B., Tobin, J., Pieter Abbeel, O., Zaremba, W.: Hindsight experience replay. *Advances in neural information processing systems* **30** (2017)
2. Andrychowicz, M., Wolski, F., Ray, A., et al.: Hindsight experience replay. In: *Advances in Neural Information Processing Systems* (2017), <https://arxiv.org/abs/1707.01495>
3. Dulac-Arnold, G., Levine, S., Mankowitz, D., et al.: Challenges of real-world reinforcement learning. *arXiv preprint arXiv:1904.12901* (2020), <https://arxiv.org/abs/1904.12901>
4. Farama Foundation: Gymnasium: A Standard Interface for Reinforcement Learning. <https://github.com/Farama-Foundation/Gymnasium> (2025), accessed: 2026-01-XX
5. Fujimoto, S., van Hoof, H., Meger, D.: Addressing function approximation error in actor-critic methods. In: *Proceedings of the 35th International Conference on Machine Learning* (2018), <https://arxiv.org/abs/1802.09477>
6. Haarnoja, T., Zhou, A., Abbeel, P., Levine, S.: Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor. In: *Proceedings of the 35th International Conference on Machine Learning* (2018), <https://arxiv.org/abs/1801.01290>
7. Henderson, P., Islam, R., Bachman, P., Pineau, J., Precup, D., Meger, D.: Deep reinforcement learning that matters. In: *Proceedings of the AAAI Conference on Artificial Intelligence* (2018), <https://arxiv.org/abs/1709.06560>
8. Kuznetsov, A., Shvechikov, P., Grishin, A., Vetrov, D.: Controlling overestimation bias with truncated mixture of continuous distributional critics. In: *Proceedings of the 37th International Conference on Machine Learning* (2020), <https://arxiv.org/abs/2005.04269>
9. Levine, S., Finn, C., Darrell, T., Abbeel, P.: End-to-end training of deep visuomotor policies. *Journal of Machine Learning Research* **17**(39), 1–40 (2016), <https://jmlr.org/papers/v17/15-522.html>
10. Levine, S., Kumar, A., Tucker, G., Fu, J.: Offline reinforcement learning: Tutorial, review, and perspectives. *arXiv preprint arXiv:2005.01643* (2020), <https://arxiv.org/abs/2005.01643>
11. Mann, H.B., Whitney, D.R.: On a test of whether one of two random variables is stochastically larger than the other. *The annals of mathematical statistics* pp. 50–60 (1947)
12. Mnih, V., Kavukcuoglu, K., Silver, D., et al.: Human-level control through deep reinforcement learning. *Nature* **518**(7540), 529–533 (2015), <https://www.nature.com/articles/nature14236>
13. Nair, A., et al.: Overcoming exploration in reinforcement learning with demonstrations. In: *Proceedings of the IEEE International Conference on Robotics and Automation* (2018), <https://arxiv.org/abs/1709.10089>
14. Plappert, M., Andrychowicz, M., Ray, A., et al.: Multi-goal reinforcement learning: Challenging robotics environments and request for research. In: *International Conference on Learning Representations* (2018), <https://arxiv.org/abs/1802.09464>

15. Rajeswaran, A., Kumar, V., Gupta, A., Levine, S.: Learning complex dexterous manipulation with deep reinforcement learning and demonstrations. In: *Robotics: Science and Systems* (2018), <https://arxiv.org/abs/1709.10087>
16. Towers, M., Kwiatkowski, A., Terry, J.U., Balis, J.U., De Cola, G., Deleu, T., Goulão, M., Kallinteris, A., Krimmel, M., Arjun, K., Perez-Vicente, R., Pierré, A., Schulhoff, S., Tai, J.J., Tan, H., Younis, O.G.: Gymnasium: A standard interface for reinforcement learning environments (2024), <https://arxiv.org/abs/2407.17032>, arXiv:2407.17032
17. Wilcoxon, F.: Individual comparisons by ranking methods. *Biometrics Bulletin* **1**(6), 80–83 (1945), <https://www.jstor.org/stable/3001968>